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# =====
# TASK 1 — SQLAlchemy Models (Baseline for Alembic)
# TASK 2 — Updated: added is_active to Student, added CourseSchedule
# =====

from sqlalchemy import (
    create_engine, Column, Integer, String, Date, Numeric,
    ForeignKey, UniqueConstraint, Boolean, Time
)
from sqlalchemy.orm import declarative_base, relationship

engine = create_engine(
    "postgresql+psycopg2://postgres:your_password@localhost:5432/college_db_orm",
    echo=True
)

Base = declarative_base()

class Department(Base):
    __tablename__ = "departments"

    dept_id = Column(Integer, primary_key=True, autoincrement=True)
    dept_name = Column(String(100), nullable=False, unique=True)
    head_of_dept = Column(String(100), nullable=False)
    location = Column(String(100))
    budget = Column(Numeric(12, 2))

    students = relationship("Student", back_populates="department")
    courses = relationship("Course", back_populates="department")
    professors = relationship("Professor", back_populates="department")

class Student(Base):
    __tablename__ = "students"

    student_id = Column(Integer, primary_key=True, autoincrement=True)
    first_name = Column(String(50), nullable=False)
    last_name = Column(String(50), nullable=False)
    email = Column(String(100), nullable=False, unique=True)
    dob = Column(Date, nullable=False)
    dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
    enrollment_year = Column(Integer)
    is_active = Column(Boolean, default=True) # added in Task 2 Step 98

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```
department = relationship("Department", back_populates="students")
enrollments = relationship("Enrollment", back_populates="student")
```

```
class Course(Base):
    __tablename__ = "courses"

    course_id = Column(Integer, primary_key=True, autoincrement=True)
    course_name = Column(String(150), nullable=False)
    credits = Column(Integer, nullable=False)
    dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
    max_seats = Column(Integer, default=60)

    department = relationship("Department", back_populates="courses")
    enrollments = relationship("Enrollment", back_populates="course")
    schedules = relationship("CourseSchedule", back_populates="course")
```

```
class Professor(Base):
    __tablename__ = "professors"

    professor_id = Column(Integer, primary_key=True, autoincrement=True)
    first_name = Column(String(50), nullable=False)
    last_name = Column(String(50), nullable=False)
    email = Column(String(100), nullable=False, unique=True)
    hire_date = Column(Date)
    dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
    salary = Column(Numeric(10, 2))

    department = relationship("Department", back_populates="professors")
```

```
class Enrollment(Base):
    __tablename__ = "enrollments"
    __table_args__ = (UniqueConstraint("student_id", "course_id"),)

    enrollment_id = Column(Integer, primary_key=True, autoincrement=True)
    student_id = Column(Integer, ForeignKey("students.student_id"), nullable=False)
    course_id = Column(Integer, ForeignKey("courses.course_id"), nullable=False)
    enrolled_on = Column(Date)
    grade = Column(String(1))

    student = relationship("Student", back_populates="enrollments")
```

```
course = relationship("Course", back_populates="enrollments")
```

```
class CourseSchedule(Base):          # added in Task 2 Step 102
    __tablename__ = "course_schedules"

    schedule_id = Column(Integer, primary_key=True, autoincrement=True)
    course_id = Column(Integer, ForeignKey("courses.course_id"), nullable=False)
    day_of_week = Column(String(10), nullable=False)
    start_time = Column(Time, nullable=False)
    end_time = Column(Time, nullable=False)
```

```
course = relationship("Course", back_populates="schedules")
```

```
if __name__ == "__main__":
    Base.metadata.create_all(engine)
    print("All tables created in college_db_orm.")
```

```
# =====
# TASK 1 — migrations/env.py
# Edit this file after running: alembic init migrations
# Replace the default env.py content with this file entirely.
# =====
```

```
from logging.config import fileConfig
from sqlalchemy import engine_from_config, pool
from alembic import context
```

```
# Import your models' Base so Alembic can compare them against the DB
import sys
import os
sys.path.insert(0, os.path.dirname(os.path.dirname(__file__)))
from models import Base
```

```
config = context.config
if config.config_file_name is not None:
    fileConfig(config.config_file_name)
```

```
target_metadata = Base.metadata
```

```
def run_migrations_offline() -> None:
    url = config.get_main_option("sqlalchemy.url")
```

```

context.configure(
    url=url,
    target_metadata=target_metadata,
    literal_binds=True,
    dialect_opts={"paramstyle": "named"},
)
with context.begin_transaction():
    context.run_migrations()

```

```

def run_migrations_online() -> None:
    connectable = engine_from_config(
        config.get_section(config.config_ini_section, {}),
        prefix="sqlalchemy.",
        poolclass=pool.NullPool,
    )
    with connectable.connect() as connection:
        context.configure(
            connection=connection,
            target_metadata=target_metadata
        )
        with context.begin_transaction():
            context.run_migrations()

```

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if context.is_offline_mode():
    run_migrations_offline()
else:
    run_migrations_online()

```

The screenshot shows a code editor with a Python migration script and its terminal output. The script defines a database engine, a declarative base, and a `Department` class. The terminal output shows the command to run the script and the resulting database schema for the `Department` table.

```

migrations > hands_on_7_models.py > ...
1 # =====
2 # TASK 1 - SQLAlchemy Models (Baseline for Alembic)
3 # TASK 2 - Updated: added is_active to Student, added CourseSchedule
4 # =====
5
6 from sqlalchemy import (
7     create_engine, Column, Integer, String, Date, Numeric,
8     ForeignKey, UniqueConstraint, Boolean, Time
9 )
10 from sqlalchemy.orm import declarative_base, relationship
11
12 engine = create_engine(
13     "postgresql+psycopg2://postgres:123456@localhost:5433/college_db_orm",
14     echo=True
15 )
16
17 Base = declarative_base()
18
19
20 class Department(Base):
21     __tablename__ = "department"
22
23     schedule_id = Column(Integer, ForeignKey("courses.schedule_id"), primary_key=True)
24     course_id = Column(Integer, ForeignKey("courses.course_id"), primary_key=True)
25     day_of_week = Column(String(10), primary_key=True)
26     start_time = Column(Time, primary_key=True)
27     end_time = Column(Time, primary_key=True)
28     PRIMARY KEY (schedule_id, course_id, day_of_week, start_time, end_time)
29     FOREIGN KEY (course_id) REFERENCES courses (course_id)
30
31
32 if __name__ == '__main__':
33     Base.metadata.create_all(engine)
34
35
36 2026-06-22 15:08:50,932 INFO sqlalchemy.engine.Engine [no key 0.00045s] {}
37 2026-06-22 15:08:50,949 INFO sqlalchemy.engine.Engine COMMIT
38 All tables created in college_db_orm.
39 PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling> python -u "c:\Users\jaswa\Downloads\Cognizant DeepSkilling\migrations\hands_on_7_models.py"

```

```
migrations > env.py > ...
1 # =====
2 # TASK 1 - migrations/env.py
3 # =====
4
5 from logging.config import fileConfig
6 from sqlalchemy import engine_from_config, pool
7 from alembic import context
8
9 # Import your models' Base so Alembic can compare them against the DB
10 import sys
11 import os
12 sys.path.insert(0, os.path.dirname(os.path.dirname(__file__)))
13 from hands_on_7_models import Base
14
15 config = context.config
16 if config.config_file_name is not None:
17     fileConfig(config.config_file_name)
18
19 target_metadata = Base.metadata
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migrations > versions > b629af09eda7_add_is_active_to_students.py > ...
1 """add is_active to students
2
3 Revision ID: b629af09eda7
4 Revises: 37ce4e2a7ec4
5 Create Date: 2026-06-22 15:15:46.228325
6
7 """
8 from typing import Sequence, Union
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10 from alembic import op
11 import sqlalchemy as sa
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```
migrations > versions > 2ba8fff342c7_add_course_schedules_table.py > ...
1  """add course_schedules table
2
3  Revision ID: 2ba8fff342c7
4  Revises: b629af09eda7
5  Create Date: 2026-06-22 15:16:33.777463
6
7  """
8  from typing import Sequence, Union
9
10 from alembic import op
11 import sqlalchemy as sa


PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling> alembic history --verbose
Revision ID: 2ba8fff342c7
Revises: b629af09eda7
Create Date: 2026-06-22 15:16:33.777463

Rev: b629af09eda7
Parent: 37ce4e2a7ec4
Path: C:\Users\jaswa\Downloads\Cognizant DeepSkilling\migrations\versions\b629af09eda7_add_is_active_to_students.py

    add is_active to students

Revision ID: b629af09eda7
Revises: 37ce4e2a7ec4
Create Date: 2026-06-22 15:15:46.228325

Rev: 37ce4e2a7ec4
Parent: <base>
Path: C:\Users\jaswa\Downloads\Cognizant DeepSkilling\migrations\versions\37ce4e2a7ec4_initial_schema.py

    initial schema

Revision ID: 37ce4e2a7ec4
Revises:
Create Date: 2026-06-22 15:14:41.173614

PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling>
```

```
PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling> alembic current
INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO [alembic.runtime.migration] Will assume transactional DDL.
2ba8fff342c7 (head)
PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling> alembic downgrade -1
INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO [alembic.runtime.migration] Will assume transactional DDL.
INFO [alembic.runtime.migration] Running downgrade 2ba8fff342c7 -> b629af09eda7, add course_schedules table
PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling> alembic downgrade -1
INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO [alembic.runtime.migration] Will assume transactional DDL.
INFO [alembic.runtime.migration] Running downgrade b629af09eda7 -> 37ce4e2a7ec4, add is_active to students
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INFO [alembic.runtime.migration] Running downgrade b629af09eda7 -> 37ce4e2a7ec4, add is_active to students
PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling> alembic downgrade base
INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO [alembic.runtime.migration] Will assume transactional DDL.
INFO [alembic.runtime.migration] Running downgrade 37ce4e2a7ec4 -> , initial schema
PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling> alembic upgrade head
INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO [alembic.runtime.migration] Will assume transactional DDL.
INFO [alembic.runtime.migration] Running upgrade -> 37ce4e2a7ec4, initial schema
INFO [alembic.runtime.migration] Running upgrade 37ce4e2a7ec4 -> b629af09eda7, add is_active to students
INFO [alembic.runtime.migration] Running upgrade b629af09eda7 -> 2ba8fff342c7, add course_schedules table
PS C:\Users\jaswa\Downloads\Cognizant DeepSkilling> alembic current
INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO [alembic.runtime.migration] Will assume transactional DDL.
2ba8fff342c7 (head)
```